Bioreactor Project & SDG Integration

Goal: To clearly articulate how my Modular Algae Bioreactor Design directly addresses and contributes to specific UN Sustainable Development Goals.

Bioreactor Function/Output	Potential SDG's
Waste Water Treatment	SDG 6, SDG 12, SDG 14
Carbon Capture	SDG 7, SDG 13
Nutrient Recovery	SDG 6, SDG 12, SDG 2
Algae Biomass Production for fertilizers and biofuels (optional)	SDG 2, SDG 7, SDG 12

Selected SDGs for Bioreactor Design

My chosen SDGs are:

- SDG 6 – Access to Clean water and Sanitation
  Connection: Wastewater treatment and nutrient recovery for water quality

- SDG 13 – Climate Action
  Connection: Carbon Capture for mitigation of greenhouse gas emissions

- SDG 12 - Responsible Consumption and Production
  Connection: Resource efficiency, waste-to-value (wastewater, nutrients, biomass).

1) SDG 6 – Access to Clean water and Sanitation

   Target 6.3 - Improve water quality, wastewater treatment, safe reuse

   - The bioreactor's wastewater treatment capability directly contributes to SDG 6 by purifying polluted water and increasing its quality for safe reuse.
   - Furthermore, its ability of nutrient recovery can prevent harmful discharge into natural water bodies, safeguarding aquatic ecosystems and promoting sustainable water resource management.

Target 6.a - International cooperation and capacity-building on water and sanitation related activities and programme, including water harvesting, desalination, water efficiency, wastewater treatment, recycling and reuse technologies.

- By developing a modular algae bioreactor technology for wastewater treatment and reuse, the project directly supports SDG 6.a, contributing to capacity-building for sustainable water management solutions that can be adapted globally

2) SDG 13 – Climate Action

Target 13.2 - Integrate climate change measures into national policies, strategies and planning.

- The modular algae bioreactor acts as a tangible climate measure, actively integrating carbon capture and efficiently removing atmospheric CO2.
- While nutrient recovery takes place, the algae undergo photosynthesis to capture CO2 efficiently.

3) SDG 12 – Responsible Consumption and Production

Target 12.2 - By 2030, achieve the sustainable management and efficient use of natural resources

- The bioreactor efficiently uses wastewater as a nutrient source and CO2 for algae growth to produce valuable biomass. This is a direct example of sustainable management and efficient use of what would otherwise be considered a waste of pollutant.

Target 12.5 - By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.

- The project directly addresses "waste generation" (wastewater is a waste stream, and CO2 from emissions can be considered waste gas) by promoting "reduction" (less pollution), "recycling" (of water and nutrients), and "reuse" (of treated water and recovered nutrients).

- The production of biomass also turns a waste input into a valuable output, preventing it from being discarded.